

OUTPUTS FOR LAB

```
y\launcher 56482 -- C:\Users\Reddy
Enter a text: hello
Enter pattern to search: hello
1. Pattern Found
```

```
y\launcher 56528 -
Enter a string: ab
Accepted
2. PS C:\Users\reddy\nlp>
```

```
3.
Original Words: ['playing', 'played', 'plays', 'player']

Stemmed Words:
playing -> play
played -> play
plays -> play
player -> player
```

```
4.
Plural Forms:
cat -> cats
bus -> buses
box -> boxes
baby -> babies
brush -> brushes
toy -> toys
```

5.

Word	Stem
playing	play
played	play
plays	play
player	player
running	run
studies	studi
studying	studi
happiness	happi

6.

```
Enter a sentence: he was playing good
Bigrams:
he -> was
was -> playing
playing -> good
```

7.

```
Enter a sentence: player plays the game
[('player', 'NN'), ('plays', 'VBZ'), ('the', 'DT'), ('game', 'NN')]
```

8.

```
Enter words: playing player plays
Assigned Tags:
playing -> JJ
player -> NN
plays -> NN
```

9.

```
Enter words: he was playing so good
he -> NN
was -> NN
playing -> VBG
so -> NN
good -> NN
```

10.

```
SA0304-NATURAL-LANGUAGE-PROCESSING\LAB\LAB10.PY'
Enter a sentence: Ram is playing
Initial Tags: [('Ram', 'NN'), ('is', 'NNS'), ('playing', 'VBG')]
After Transformation Rules: [('Ram', 'NNP'), ('is', 'NNS'), ('playing', 'VBG'
)]
PS C:\Users\reddy\nlp>
```

11.

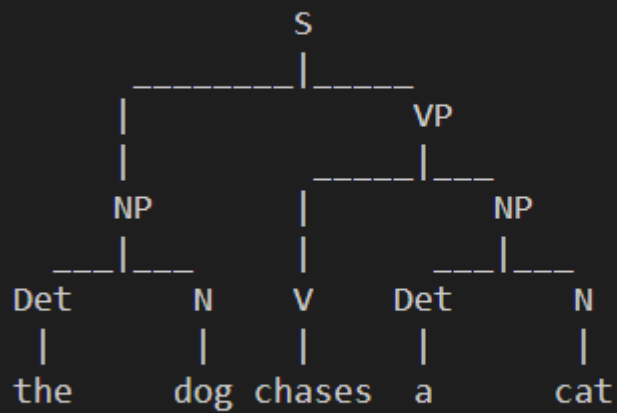
```
LANGUAGE-PROCESSING\LAB\LAB11.PY'
Enter a sentence: the dog chases a cat
Sentence is Valid (Parsed Successfully)
PS C:\Users\reddy\nlp>
```

```
LANGUAGE-PROCESSING\LAB\LAB12.PY'
Enter a sentence: the dog chases a cat
Sentence is Valid (Parsed Successfully)
PS C:\Users\reddy\nlp>
```

12.

```
LANGUAGE-PROCESSING\LAB\LAB13.PY
Enter a sentence: the dog chases a cat
```

Parse Trees:



```
PS C:\Users\reddy\nlp>
```

13.

```
LANGUAGE-PROCESSING\LAB\LAB14.PY
Enter a sentence: some dogs run
Sentence has Valid Agreement (Grammatically Correct)
PS C:\Users\reddy\nlp>
```

14.

```
Enter a sentence: the dog chases a cat

Most Probable Parse Tree(s):
(S
  (NP (Det the) (N dog))
  (VP (V chases) (NP (Det a) (N cat)))) (p=0.00217728)
Probability: 0.0021772799999999998
PS C:\Users\reddy\nlp>
```

15.

16.

```
LANGUAGE-PROCESSING\LAB\LAB16.PY
```

```
Named Entities:
```

```
Entity: Elon Musk
```

```
Label: PERSON
```

```
Explanation: People, including fictional
```

```
Entity: Tesla
```

```
Label: ORG
```

```
Explanation: Companies, agencies, institutions, etc.
```

```
Entity: South Africa
```

```
Label: GPE
```

```
Explanation: Countries, cities, states
```

```
PS C:\Users\reddy\nlp> 
```

17.

```

...
Name: bank.v.03
Definition: do business with a bank or keep an account
at a bank
Examples: ['Where do you bank in this town?']

Name: bank.v.04
Definition: act as the banker in a game or in gambling
Examples: []

Name: bank.v.05
Definition: be in the banking business
Examples: []

Name: deposit.v.02
Definition: put into a bank account
Examples: ['She deposits her paycheck every month']

Name: bank.v.07
Definition: cover with ashes so to control the rate of
burning
Examples: ['bank a fire']

Name: trust.v.01
Definition: have confidence or faith in
Examples: ['We can trust in God', 'Rely on your friends
', 'bank on your good education', "I swear by my grandm
other's recipes"]
PS C:\Users\reddy\nlp>

```

18.

```

LANGUAGE-PROCESSING\LAB\LAB18.PY
Enter a logical expression: Loves(John,Mary)
Invalid FOPC expression
PS C:\Users\reddy\nlp>

```

19.

```

Word: bank
Sense: depository_financial_institution.n.01
Definition: a financial institution that accepts deposi
ts and channels the money into lending activities
PS C:\Users\reddy\nlp>

```

20.

```
Enter your search query: natural language processing

Document Ranking:

Rank: 1
Score: 0.581
Document: Natural language processing is a branch of artificial intelligence

Rank: 2
Score: 0.562
Document: Python is widely used for natural language processing

Rank: 3
Score: 0.0
Document: Machine learning is used in artificial intelligence

Rank: 4
Score: 0.0
Document: Information retrieval helps find relevant documents

PS C:\Users\reddy\nlp> 
```

```
LANGUAGE-PROCESSING\LAB\LAB21.PY
Sentence: The intelligent student reads the book in the
library

Noun Phrases and Meanings:

Noun Phrase: student
Meaning: a person who studies

Noun Phrase: book
Meaning: a collection of written pages

Noun Phrase: library
Meaning: a place containing books

PS C:\Users\reddy\nlp>
```

22.

```
LANGUAGE-PROCESSING\LAB\LAB22.PY
Text: Ravi went to the market. He bought some fruits. He
returned home.

Reference Resolution:

He refers to Ravi
He refers to Ravi
PS C:\Users\reddy\nlp>
```

23.

```
LANGUAGE-PROCESSING\LAB\LAB23.PY
Text Coherence Analysis:

Sentence 1 and Sentence 2
Common words: {'processing', 'natural', 'language'}
Coherence score: 3

Sentence 2 and Sentence 3
Common words: {'language'}
Coherence score: 1

Total Coherence Score: 4
The text is coherent.
PS C:\Users\reddy\nlp>
```


24.

```
LANGUAGE-PROCESSING\LAB\LAB24.PY
Enter a sentence: Hello, how are you?
Dialog Act: Greeting
PS C:\Users\reddy\nlp>
```

25.

```
Enter prompt: write two lines about artificial inetelligence
Prompt: write two lines about artificial inetelligence
Generated Text: Technology makes our life easier.
```

26.

```
Enter English text: i love learning python
French: J'adore apprendre le python
```